文件: comm.kicad_sch

COMM

文件: usb.kicad_sch

USB

文件: io.kicad_sch

IO

文件: power.kicad_sch

Power

文件: core.kicad_sch

Core

文件: display.kicad_sch

Display

Sheets: Power / Core / Display / COMM / USB / IO. Cross-sheet wiring via global labels.

ZHISHU HMI Control Board — T113-S3 — Rev 1.0 — 2026-08-19

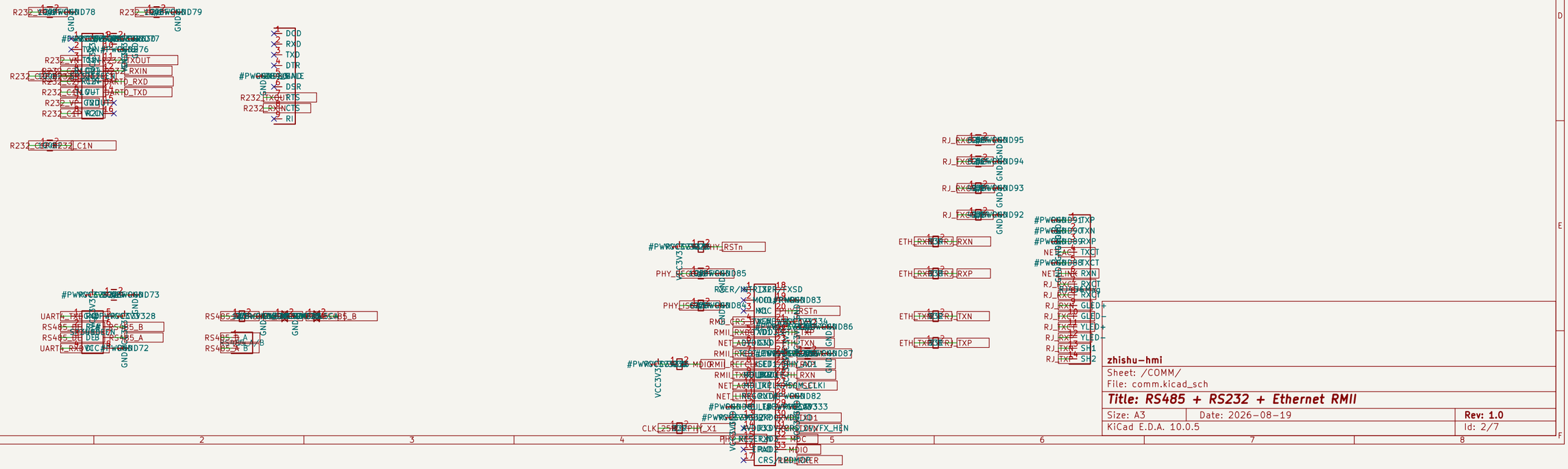
zhishu-hmi

Sheet: /

File: hmi-board.kicad_sch

Title: HMI Control Board T113-S3

Size: A3	Date: 2026-08-19	**Rev: 1.0**
KiCad E.D.A. 10.0.5		Id: 1/7



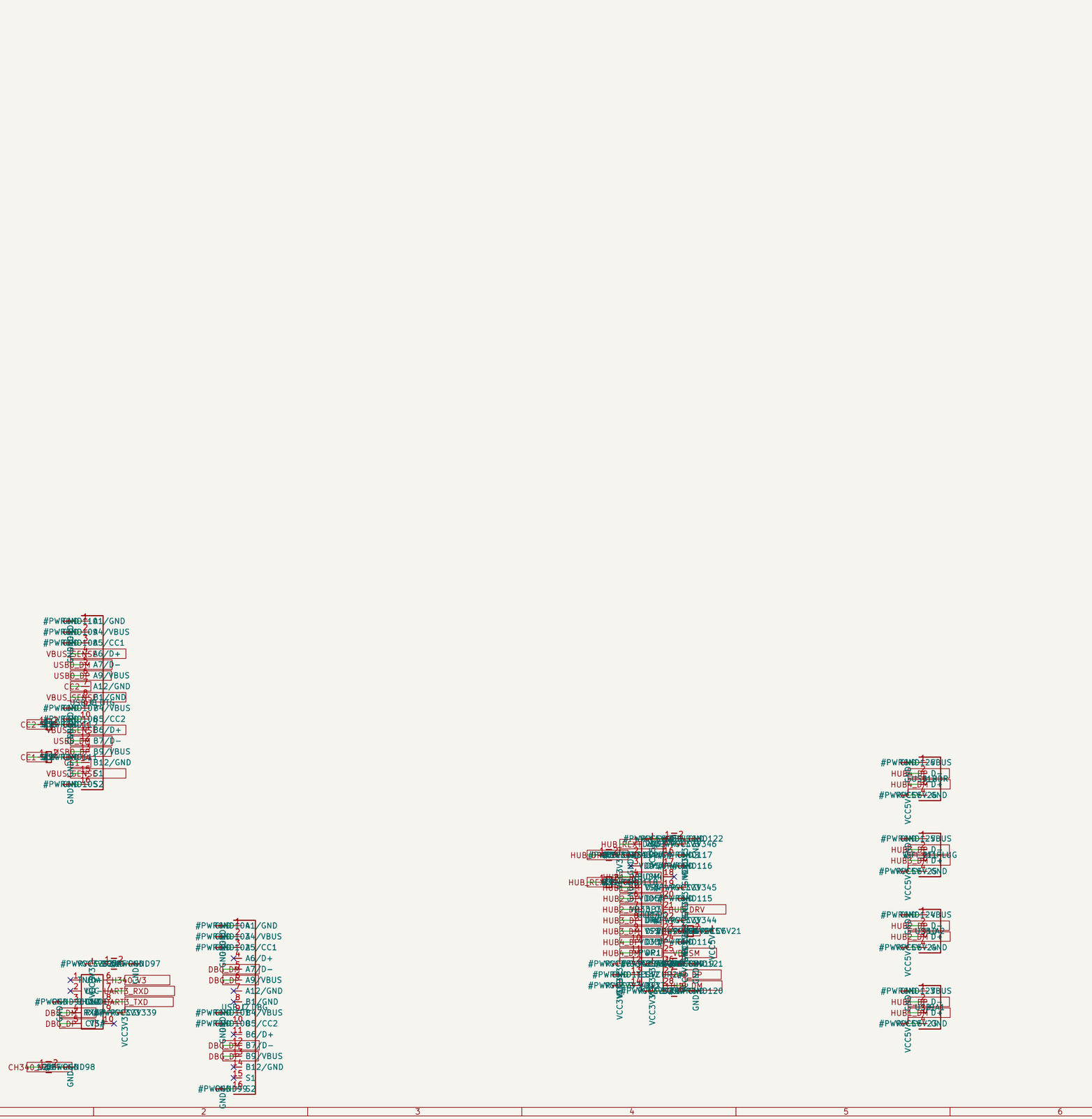
zhishu-hmi	
Sheet: /COMM/	
File: comm.kicad_sch	
Title: RS485 + RS232 + Ethernet RMII	
Size: A3	Date: 2026-08-19
KiCad E.D.A. 10.0.5	Rev: 1.0
	Id: 2/7

SY8089 Vout=0.6*(1+Rup/Rdn): 3V3=150K/33K, 1V5=150K/100K, 0V9=100K/100K, 1V8=150K/68K

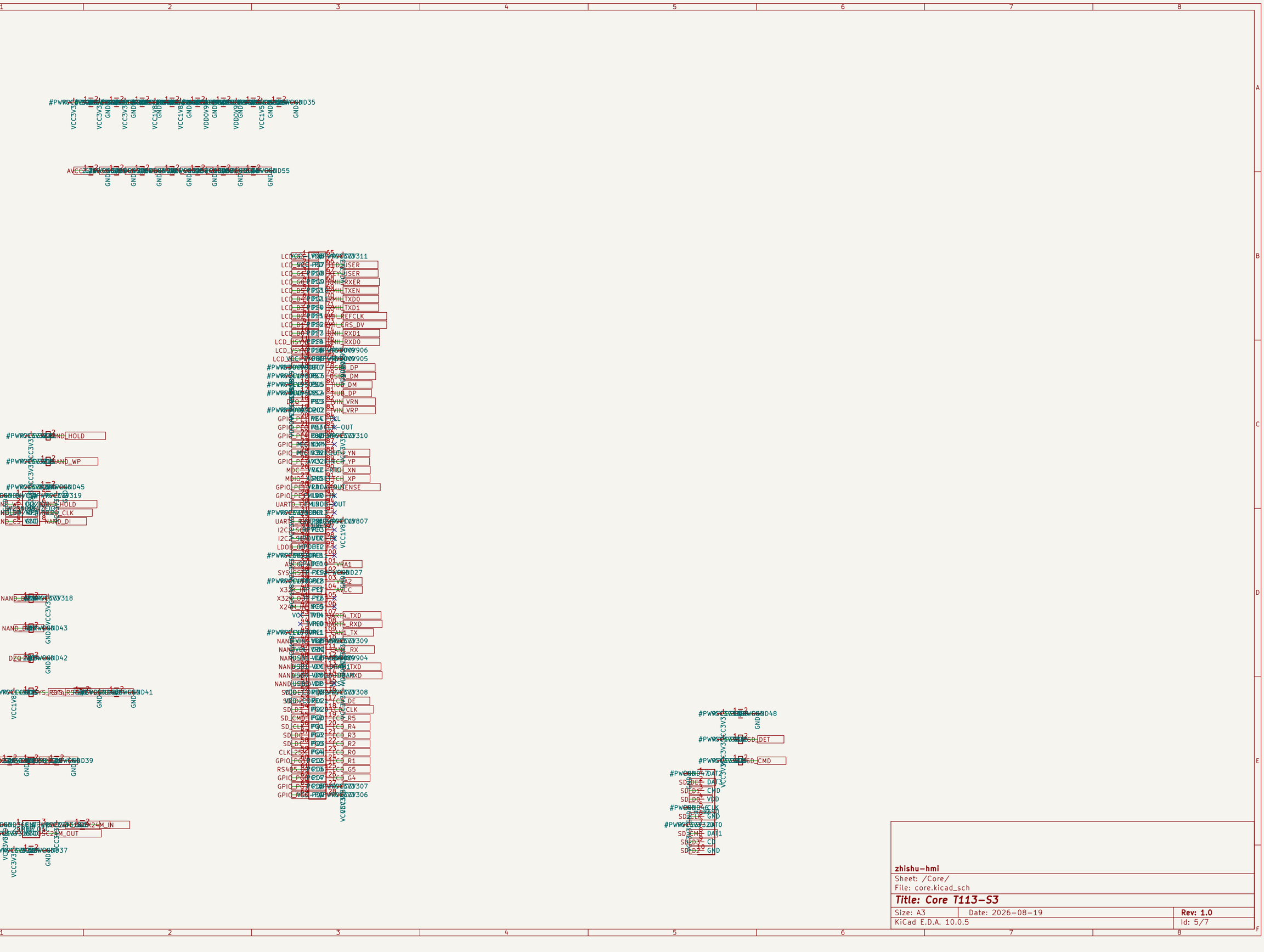
DC24V -> TPS5430 (5V) -> SY8089 x3 (3V3/1V5/0V9) + XC6206 (1V8)

HMI Control Board — Power Tree

zhishu-hmi		
Sheet: /Power/		
File: power.kicad_sch		
Title: Power 24V->5V/3V3/1V5/0V9/1V8		
Size: A3	Date: 2026-08-19	**Rev: 1.0**
KiCad E.D.A. 10.0.5		Id: 3/7



Hub downstream: USB-A x2, WIFI/BT plug-in header, spare header. (SDIO WiFi conflicts with RMII on PG0-PG5)


LCD_B7
LCD_B6
LCD_B5
LCD_B4
LCD_B3
LCD_B2
LCD_B1
LCD_B0
LCD_HSYNC
LCD_VSYNC
LCD_DE
LCD_CLK
LCD_R5
LCD_R4
LCD_R3
LCD_R2
LCD_R1
LCD_R0
LCD_G5
LCD_G4
LCD_G3
LCD_G2
LCD_G1
LCD_G0
LCD_B5
LCD_B4
LCD_B3
LCD_B2
LCD_B1
LCD_B0
GPIO_PE0
GPIO_PE1
GPIO_PE2
GPIO_PE3
GPIO_PE4
GPIO_PE5
GPIO_PE6
GPIO_PE7
GPIO_PE8
GPIO_PE9
GPIO_PE10
GPIO_PE11
GPIO_PE12
GPIO_PE13
GPIO_PE14
GPIO_PE15
GPIO_PE16
GPIO_PE17
GPIO_PE18
GPIO_PE19
GPIO_PE20
GPIO_PE21
GPIO_PE22
GPIO_PE23
GPIO_PE24
GPIO_PE25
GPIO_PE26
GPIO_PE27
GPIO_PE28
GPIO_PE29
GPIO_PE30
GPIO_PE31
UART4_TXD
UART4_RXD
CAN1_TX
CAN1_RX
RGMII_TXD1
RGMII_TXD0
RGMII_TXCTL
RGMII_RXD1
RGMII_RXD0
RGMII_RXCTL
RGMII_TXD3
RGMII_TXD2
RGMII_RXD3
RGMII_RXD2
RGMII_TXCLK
RGMII_RXCLK
RGMII_MDC
RGMII_MDIO
RGMII_REFCLK
RGMII_CRS_DV
USB_DP
USB_DM
USB_DM
USB_DP
SD_DET
SD_CMD
SD_CLK
SD_DAT0
SD_DAT1
SD_DAT2
SD_DAT3
NAND_WP
NAND_HOLD
NAND_CLK
NAND_DI
NAND_DO
NAND_CS
X32K_OUT
X32K_IN
X24M_IN
X24M_OUT
RS485_DE
RS485_RX
RS485_TX
I2C_SCL
I2C_SDA
SYS_RST
VCC3V3
VCC1V8
GND
zhishu-hmi
Sheet: /Core/
File: core.kicad_sch
Title: Core T113-S3
Size: A3
Date: 2026-08-19
Rev: 1.0
KiCad E.D.A. 10.0.5
Id: 5/7

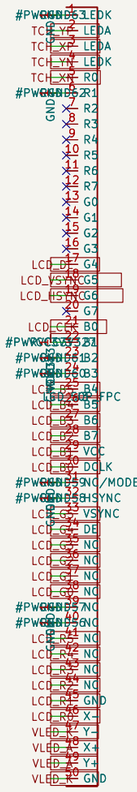


zhishu-hmi

Sheet: /IO/
File: io.kicad_sch

Title: IO: LED/KEY/Expansion/VIN sense

Size: A3	Date: 2026-08-19	**Rev: 1.0**
KiCad E.D.A. 10.0.5		Id: 6/7



zhishu-hmi

Sheet: /Display/

File: display.kicad_sch

Title: Display LCD50P + Backlight + Touch

Size: A3	Date: 2026-08-19	**Rev: 1.0**
KiCad E.D.A. 10.0.5		Id: 7/7